

# Examination Procedure for 6.1: Virtual Reality and Optimization in Environmental Engineering

## Summer Semester 2026

### 1. Overview

The assessment for this course is based on a group project. Students will work in **groups of two** throughout the semester.

Each group is expected to develop its own project idea within the thematic scope of the course. Project ideas will be discussed during the lecture period, and guidance will be provided to help refine the topic and define a suitable project plan.

The deadline for group formation and finalization of the project topic is **30 June 2026**.

Detailed instructions and further guidance will be provided during the course and shared via AULIS.

### 2. Project Work

Students are expected to work on a practical project that is implemented in **Python** and addresses a topic related to **environmental engineering**. The project topic should be proposed by the group and then discussed with the instructor during the semester.

The aim of the project is to apply methods and tools introduced in the course in a structured and practical way. Support will be provided during the lecture weeks to help students develop their ideas and make steady progress.

### 3. Use of GitHub

The use of a **GitHub** repository is recommended for managing code, data, and other project materials in an organized way.

GitHub can support collaboration within the group and help document the development process of the project. Since this may be new for you, detailed instruction videos have been prepared. For more details, refer to VRO\_Setup\_Videos\_Instructions.pdf, which is uploaded in the 03 - Student Project Resources folder in AULIS.

No prior experience with GitHub is required.

### 4. Development Environment / IDE

Students may use any development environment (IDE) of their choice for the project.

However, **Visual Studio Code (VS Code)** is recommended, as it is well-suited for Python-based project work and will be used in the course demonstrations. For more details, refer to VRO\_Setup\_Videos\_Instructions.pdf, which is uploaded in the 03 - Student Project Resources folder in AULIS.

No prior experience with VS Code is required.

## 5. Final Submission

At the end of the course, each group must submit a written report, give a presentation, and provide the project materials in a well-organized form.

The written report should be approximately **20 pages** in length and must include bibliography/references. It should clearly present the project topic and motivation, the methodology and implementation, the results, and the discussion and conclusions.

The submission deadline for the written report is on **1 September 2026**. Submission may take place via **AULIS** and/or **GitHub**.

Each group will also give a final project presentation. The presentation time is **15 minutes** and will be followed by a **10-minute discussion**.

The presentations will take place on **25 September 2026**.

In addition, each group is expected to maintain a GitHub repository containing the relevant source code, datasets where applicable, and a short description or documentation of the project.

## 6. Additional Information

Further details regarding the project scope, expectations, and formal requirements will be discussed during the lecture and made available on AULIS.

Since several tools and workflows used in the project may be new to students, the necessary introductions will be provided as part of the course. The intention is to support students step by step during the semester.